PROBLEM-2 Analysis Report

Satisfying configuration (Greedy Best First Search)

Search algorithm: Greedy best-first search
Heuristic: FF(ff()) (Fast Forward)
Finds a valid plan if the search succeeds, but does not guarantee an optimal plan.
Fast forward heuristic may not be admissible.

Sequence of actions:
move rover base sitea (1)
collect-soft rover rock1 bay1 sitea (1)
move rover sitea base (1)
deposit rover bay1 rock1 base (1)
move rover base siteb (1)
move rover siteb sited (1)
move rover sited sitee (1)
pick-drill rover drill1 sitee (1)
move rover sitee sited (1)
collect-soft rover rock4 bay1 sited (1)
move rover sited siteb (1)
move rover siteb base (1)
deposit rover bay1 rock4 base (1)
move rover base siteb (1)
collect-soft rover rock2 bay2 siteb (1)
move rover siteb base (1)
deposit rover bay2 rock2 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
collect-hard rover rock3 bay1 sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock3 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
move rover sitee sitef (1)
collect-hard rover rock5 bay1 sitef (1)
move rover sitef sitee (1)

move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock5 base (1)

Plan length = total number of actions : 33 step(s).
Plan cost: 33
Expanded 68 state(s)

Planner time: 0.16s

Verification using VAL, fastforward

```
Solution found.
Peak memory: 11620 KB
Remove intermediate file output.sas
search exit code: 0

INFO     Running validate.
INFO     validate stdin: None
INFO     validate time limit: None
INFO     validate memory limit: None
INFO     validate command line string: /mnt/76BA9F67BA9F22A3/d1/sem5/AI/labs/VAL/validate ../grp_assignment_2/domain.pddl ../g
rp_assignment_2/problem.pddl sas_plan
Checking plan: sas_plan
Plan executed successfully - checking goal
Plan valid
Final value: 33

Successful plans:
Value: 33
 sas_plan 33

validate exit code: 0

INFO     Planner time: 0.15s
```

Optimal Configuration (A* Search)

Search algorithm: A*
Heuristic: LM-cut (lmcut()) landmark cut
Optimal plan, provided the task has nonnegative action costs and the search completes successfully. LM-cut is admissible

move rover base siteb (1)
collect-soft rover rock2 bay1 siteb (1)
move rover siteb sited (1)
collect-soft rover rock4 bay2 sited (1)
move rover sited siteb (1)
move rover siteb base (1)
deposit rover bay1 rock2 base (1)
deposit rover bay2 rock4 base (1)

move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
pick-drill rover drill1 sitee (1)
move rover sitee sitef (1)
collect-hard rover rock5 bay1 sitef (1)
move rover sitef sitee (1)
move rover sitee sitec (1)
collect-hard rover rock3 bay2 sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock5 base (1)
deposit rover bay2 rock3 base (1)
move rover base sitea (1)
collect-soft rover rock1 bay1 sitea (1)
move rover sitea base (1)
deposit rover bay1 rock1 base (1)

Plan length: 25 step(s).
Plan cost: 25
Expanded 5087 state(s)
Planner time: 0.59s

Verified with VAL, fastforward

```
Peak memory: 12172 KB                                                                                                [0/1989]
Remove intermediate file output.sas
search exit code: 0

INFO     Running validate.
INFO     validate stdin: None
INFO     validate time limit: None
INFO     validate memory limit: None
INFO     validate command line string: /mnt/76BA9F67BA9F22A3/d1/sem5/AI/labs/VAL/validate ../grp_assignment_2/domain.pddl ../g
rp_assignment_2/problem.pddl sas_plan
Checking plan: sas_plan
Plan executed successfully - checking goal
Plan valid
Final value: 25

Successful plans:
Value: 25
 sas_plan 25

validate exit code: 0

INFO     Planner time: 0.60s
```

Extended Scenario

Satisfying Configuration (Greedy Best First Search)

Search algorithm: Greedy best-first search
Heuristic: FF(ff()) Fast Forward which uses relaxed graph
Finds a valid plan if the search succeeds, but does not guarantee an optimal plan.
FF is not generally admissible

Sequence of actions:

move rover base sitea (1)
collect-soft rover rock1 bay1 sitea (1)
move rover sitea base (1)
deposit rover bay1 rock1 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
pick-drill rover drill1 sitee (1)
move rover sitee sitec (1)
collect-hard rover rock3 bay1 sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock3 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sited (1)
move rover sited siteb (1)
collect-soft rover rock2 bay1 siteb (1)
move rover siteb sited (1)
collect-soft rover rock4 bay2 sited (1)
move rover sited sitee (1)

move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock2 base (1)
deposit rover bay2 rock4 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
move rover sitee sitef (1)
collect-hard rover rock5 bay1 sitef (1)
collect-hard rover rock7 bay2 sitef (1)
move rover sitef sitee (1)
move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock5 base (1)
deposit rover bay2 rock7 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
collect-soft rover rock6 bay1 sitee (1)
move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock6 base (1)

Plan length = total number of actions : 46 step(s).
Plan cost: 46
Expanded 141 state(s)

Planner time: 0.16s

Verification using VAL, fastforward



Optimal Configuration (A* search)

Search algorithm: A*
Heuristic: LM-cut (lmcut())
Optimal plan, provided the task has nonnegative action costs and the search completes successfully. LM-cut is admissible

Sequence Actions:

move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sited (1)
collect-soft rover rock4 bay1 sited (1)
move rover sited siteb (1)
collect-soft rover rock2 bay2 siteb (1)
move rover siteb sited (1)
move rover sited sitee (1)
pick-drill rover drill1 sitee (1)
move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock4 base (1)
deposit rover bay2 rock2 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
move rover sitec sitee (1)
move rover sitee sitef (1)
collect-hard rover rock5 bay1 sitef (1)
collect-hard rover rock7 bay2 sitef (1)

move rover sitef sitee (1)
move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock5 base (1)
deposit rover bay2 rock7 base (1)
move rover base sitea (1)
move rover sitea sitec (1)
collect-hard rover rock3 bay1 sitec (1)
move rover sitec sitee (1)
collect-soft rover rock6 bay2 sitee (1)
move rover sitee sitec (1)
move rover sitec sitea (1)
move rover sitea base (1)
deposit rover bay1 rock3 base (1)
deposit rover bay2 rock6 base (1)
move rover base sitea (1)
collect-soft rover rock1 bay1 sitea (1)
move rover sitea base (1)
deposit rover bay1 rock1 base (1)

Plan length = total number of actions : 40 step(s).
Plan cost: 40
Expanded 2230074 state(s)

Planner time: 127.40s

Verification using VAL, fastforward

```
Solution found.
Peak memory: 207720 KB
Remove intermediate file output.sas
search exit code: 0

INFO    Running validate.
INFO    validate stdin: None
INFO    validate time limit: None
INFO    validate memory limit: None
INFO    validate command line string: /mnt/76BA9F67BA9F22A3/d1/sem5/AI/labs/VAL/validate ../grp_assignment_2/domain.pddl ../grp_assignment_2/problem_extended.pddl sas_plan
Checking plan: sas_plan
Plan executed successfully - checking goal
Plan valid
Final value: 40

Successful plans:
Value: 40
 sas_plan 40
validate exit code: 0

INFO    Planner time: 126.78s
```

Plan Analysis

1. Comparison of plan lengths
Normal scenario: GBFS 33 steps, A* 25 steps.
Extended scenario: GBFS 46 steps, A* 40 steps.

GBFS returns a plan almost immediately, expanding only 68 and 141 states in 0.16s, but FF is not admissible so it keeps the first plan it finds. It commits to whichever rock looks closest at each step, which leaves several single-rock round trips to Base. A* with LM-cut is admissible, so its plan is the shortest possible, but it has to expand far more states to prove that: 5087 states in 0.59s for the normal scenario and 2230074 states in 127.40s for the extended one. The extra 8 and 6 steps in the GBFS plans are the price of that speed.

2. Use of the two storage bays and the drill
The optimal plan fills both bays before returning whenever two rocks are reachable on the same leg. In the normal scenario it loads rock2 into bay1 and rock4 into bay2 on the SiteB/SiteD leg and deposits both at Base, then loads rock5 into bay1 and rock3 into bay2 on the SiteF/SiteC leg and deposits both, and only rock1 is carried on its own at the end. That is three return trips instead of five.

The drill is picked up at step 12, at SiteE, while the rover is already on its way to the hard rocks, so no separate trip is made for it. Both hard rocks, rock3 and rock5, are collected after that step, which is required because collect-hard needs drill_picked. In the extended plan the pickup happens at step 9, again at SiteE on a leg the rover was already making.

3. Reaching SiteB and SiteD after the dust storm
With Base <-> SiteB closed, SiteB has only one remaining neighbour, SiteD, so every route to SiteB now runs through SiteD. SiteD itself can be entered from SiteC or from SiteE. The optimal plan takes Base -> SiteA -> SiteC -> SiteD -> SiteB, so the one-way route SiteC -> SiteD is used. It is not the only way in, since Base -> SiteA -> SiteC -> SiteE -> SiteD -> SiteB also works, but that route is one move longer, which is why the planner prefers the one-way slope.

The way back is different, because SiteD -> SiteC does not exist. The rover returns SiteB -> SiteD -> SiteE -> SiteC -> SiteA -> Base, looping round through SiteE. Along with the two extra rocks, this detour is what makes the extended plan longer: a trip to SiteB that used to cost one move each way now costs four moves in and five back.